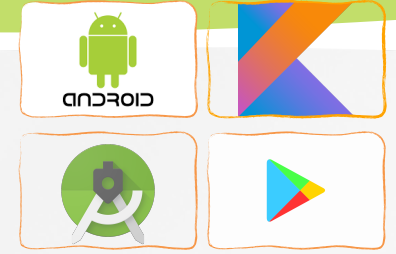




# Software Development for Mobile Devices

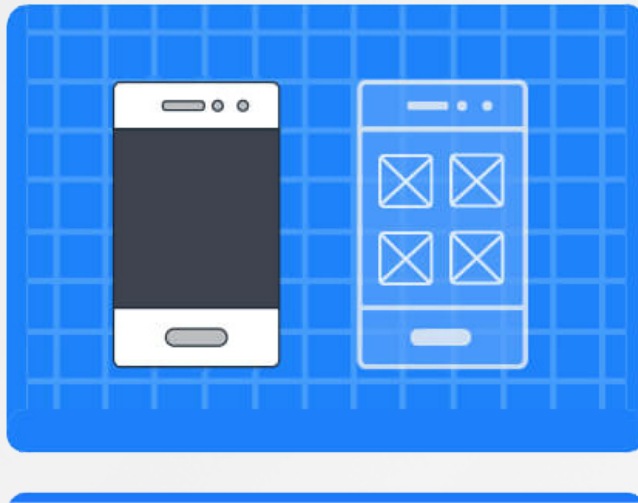
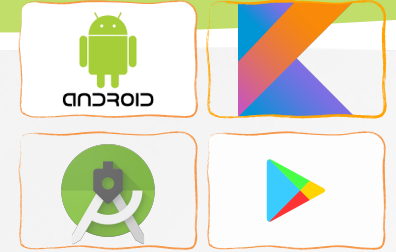
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# Content



- » Android OS
- » Hardware and software
- » IDEs
- » Android Virtual Device Manager (AVD)
- » Manifest file
- » Currency converter app
- » Mobile market

# Demo



 **Kotlin**

# Example1: Currency converter



Implementing a currency converter:

USD  $\rightarrow$  Euro  $\rightarrow$  Colon (CR)

Note.

Naive implementation using a fixed exchange rate:

1 Costa Rican Colon = 0.0019 U.S. dollars 1 Euro = 1.35  
U.S. dollars

»

## Example1: Currency converter (cont.)



» Implementing a currency converter:

USD → VND → CAD

» Note: we should use the fixed exchange rate:

- 1 usd = 23172.5 Vietnamese Dong
- 1 usd = 1.31 Canadian Dollar

# Example1: Currency converter (cont.)



```
package com.dnd.a02_kotlin
import android.content.Intent
import android.net.Uri
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import java.text.DecimalFormat

class MainActivity : AppCompatActivity() {
    val usaDf = DecimalFormat("###,###,###,###.##");
    val usd2cad = 1.32
    val usd2vnd = 23172.50

    var convertBtn : Button? = null
    var clearBtn : Button? = null
    var usdET : EditText? = null
    var cadET : EditText? = null
    var vndET : EditText? = null
```

# Example1: Currency converter (cont.)



```
override fun onCreate(savedInstanceState: Bundle?) {  
    super.onCreate(savedInstanceState)  
    setContentView(R.layout.activity_main)  
  
    convertBtn = findViewById(R.id.btnConvert) as Button  
    clearBtn = findViewById(R.id.btnClear) as Button  
    usdET = findViewById(R.id.etUSD) as EditText  
    cadET = findViewById(R.id.etCAD) as EditText  
    vndET = findViewById(R.id.etVND) as EditText
```

# Example1: Currency converter (cont.)



```
convertBtn!!.setOnClickListener {  
    val usdVal = usdET!!.text.toString().toDouble()  
    val cadVal = usdVal * usd2cad  
    val vndVal = usdVal * usd2vnd  
  
    cadET!!.setText(usaDf.format(cadVal))  
    vndET!!.setText(usaDf.format(vndVal))  
}  
  
clearBtn!!.setOnClickListener {  
    cadET!!.setText("")  
    cadET!!.setText("")  
    vndET!!.setText("")  
}  
}
```



# Example1: Currency converter (cont.)



```
convertBtn!!.setOnClickListener {  
    val usdVal = usdET!!.text.toString().toDouble()  
    val cadVal = usdVal * usd2cad  
    val vndVal = usdVal * usd2vnd  
  
    cadET!!.setText(usaDf.format(cadVal))  
    vndET!!.setText(usaDf.format(vndVal))  
}  
  
clearBtn!!.setOnClickListener {  
    cadET!!.setText("")  
    cadET!!.setText("")  
    vndET!!.setText("")  
}  
}
```

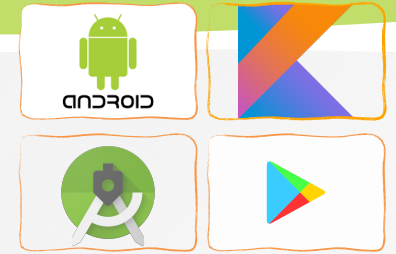
A02-Kotlin

Currency Converter

|     |         |
|-----|---------|
| USD | 10      |
| CAD | 13.2    |
| VND | 231,725 |

CLEAR CONVERT

# Q&A





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